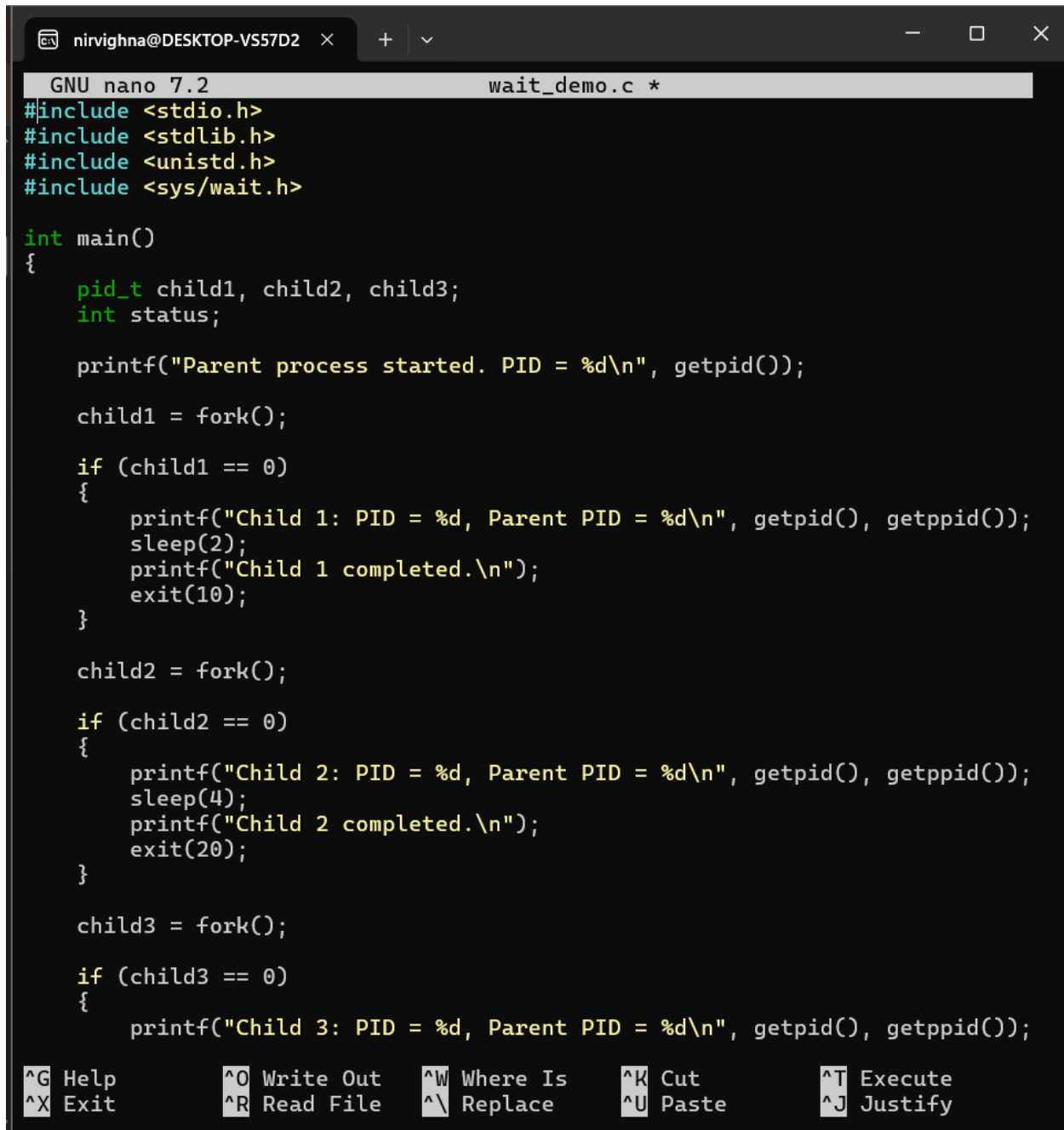


PRACTICAL -4:



```
GNU nano 7.2 wait_demo.c *
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    pid_t child1, child2, child3;
    int status;

    printf("Parent process started. PID = %d\n", getpid());

    child1 = fork();

    if (child1 == 0)
    {
        printf("Child 1: PID = %d, Parent PID = %d\n", getpid(), getppid());
        sleep(2);
        printf("Child 1 completed.\n");
        exit(10);
    }

    child2 = fork();

    if (child2 == 0)
    {
        printf("Child 2: PID = %d, Parent PID = %d\n", getpid(), getppid());
        sleep(4);
        printf("Child 2 completed.\n");
        exit(20);
    }

    child3 = fork();

    if (child3 == 0)
    {
        printf("Child 3: PID = %d, Parent PID = %d\n", getpid(), getppid());
    }
}
```

 ^G Help	 ^O Write Out	 ^W Where Is	 ^K Cut	 ^T Execute
 ^X Exit	 ^R Read File	 ^I Replace	 ^U Paste	 ^J Justify

```
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ ./wait_demo
Parent process started. PID = 3616
Child 1: PID = 3617, Parent PID = 3616
Child 2: PID = 3618, Parent PID = 3616

Parent waiting for children using wait()...
Child 3: PID = 3619, Parent PID = 3616
Child 3 completed.
wait(): Child PID 3619 completed.

Parent waiting for remaining children using waitpid()...
Child 1 completed.
waitpid(): Child 1 completed.
Child 2 completed.
waitpid(): Child 2 completed.
waitpid(): Child 3 completed.

All child processes completed.
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ |
```

```
nirvighna@DESKTOP-VS57D2  ×  +  ▾
GNU nano 7.2                zombie.c *
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>

int main()
{
    pid_t pid;

    pid = fork();

    if (pid < 0)
    {
        perror("fork failed");
        return 1;
    }

    if (pid == 0)
    {
        printf("Child process started. PID = %d\n", getpid());
        printf("Child process exiting...\n");
        exit(0);
    }
    else
    {
        printf("Parent process started. PID = %d\n", getpid());
        printf("Child PID = %d\n", pid);

        printf("Parent sleeping for 30 seconds...\n");

        sleep(30);

        printf("Parent exiting...\n");
    }

    return 0;
}

^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify
```

```
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ ./zombie
Parent process started. PID = 3657
Child PID = 3658
Parent sleeping for 30 seconds...
Child process started. PID = 3658
Child process exiting...
```

```

nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ ps -el
 F S      UID        PID      PPID    C  PRI   NI     ADDR  SZ  WCHAN  TTY          TIME CMD
 4 S      0           1          0    0   80    0      - 5527  -      ?           00:00:04 system
 5 S      0           2          1    0   80    0      - 795   -      hvc0         00:00:01 init-s
 0 S      0           8          2    0   80    0      - 799   -      hvc0         00:00:00 init
 4 S      0          54          1    0   79   -1     - 16710 -      ?           00:00:15 system
 4 S      0         103          1    0   80    0      - 6285  -      ?           00:00:05 system
 4 S     991         156          1    0   80    0      - 5338  -      ?           00:00:00 system
 4 S     996         157          1    0   80    0      - 22761 -      ?           00:00:01 system
 4 S      0         166          1    0   80    0      - 1062  -      ?           00:00:00 cron
 4 S     101         167          1    0   80    0      - 2442  -      ?           00:00:02 dbus-d
 4 S      0         184          1    0   80    0      - 4497  -      ?           00:00:00 system
 4 S      0         186          1    0   80    0      - 439027 -      ?           00:00:01 wsl-pr
 4 S     102         190          1    0   80    0      - 55631 -      ?           00:00:01 rsyslo
 4 S      0         204          1    0   80    0      - 782   -      tty1         00:00:00 agetty
 4 S      0         226          1    0   80    0      - 26760 -      ?           00:00:00 unatte
 4 S      0         317          2    0   80    0      - 1676  -      pts/1        00:00:00 login
 4 S    1000         373          1    0   80    0      - 5084  -      ?           00:00:00 system
 5 S    1000         376        373    0   80    0      - 5295  -      ?           00:00:00 (sd-pa
 4 S    1000         402        317    0   80    0      - 1521  poll_s pts/1        00:00:00 bash
 4 S     990        1231          1    0   80    0      - 77045 -      ?           00:00:00 polkit
 5 S      0        2791          2    0   80    0      - 796   -      ?           00:00:00 Sessio
 5 S      0        2792        2791    0   80    0      - 800   -      ?           00:00:00 Relay(
 4 S    1000        2793        2792    0   80    0      - 1521  do_wai pts/0        00:00:00 bash
 5 S      0        3663         103    0   80    0      - 6286  -      ?           00:00:00 (udev-
 5 S      0        3664         103    0   80    0      - 6286  -      ?           00:00:00 (udev-
 0 R    1000        3665        2793   99   80    0      - 2082  -      pts/0        00:00:00 ps
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$

```